CURRICULUM VITAE

Zehang Li, PhD

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WORK EXPERIENCE

Jul 2023 – now **Assistant Professor**

- College of Health Science and Technology, Shanghai Jiao Tong University School of Medicine, Shanghai, China
- Department of Radiology, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

Oct 2021 – Jul 2023 **Medical Imaging Engineer**

- Shanghai Pulse Medical Inc, Shanghai, China

EDUCATIONAL BACKGROUND

Dec 2019 – Jan 2021 **Joint PhD in Cardiovascular Imaging**

- Division of Cardiology, Johns Hopkins Hospital, Johns Hopkins University School of Medicine, Maryland, United States

Sep 2016 – Sep 2021 **PhD in Biomedical Engineering**

- School of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai, China

Mar 2016 – May 2016 **Exchange Student in Medical Imaging**

- School of Technology and Health (STH), KTH Royal Institute of Technology, Stockholm, Sweden

Sep 2012 – Jun 2016 **Bachelor in Biomedical Engineering**

- School of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai, China

RESEARCH EXPERIENCE

My main area of interest is cardiovascular imaging and non-invasive assessment of coronary artery physiology. The research aims to investigate the underlying mechanisms of myocardial ischemia and identify indices that can guide the implementation of novel therapeutic techniques to improve the prognosis of patients with ischemic heart disease. Specifically, the main research content aims to develop methods for automatic segmentation and plaque quantification of coronary arteries and fast fractional flow reserve calculation algorithms in coronary CT angiography images.

I have participated in the study design, implementation, and subsequent analysis of several clinical studies, including FAVOR II China, FAVOR III China, and CORE320 trials.

GRANTS (as the Principal Applicant)

- 2024 Shanghai Science and Technology Development Fund, 2024/7-2027/6 (RMB 200,000)
- 2024 Fundamental Research Fund for the Central Universities at Shanghai Jiao Tong University, 2024/1-2026/12 (RMB 200,000)
- 2023 Scientific Research Fund for the Young Faculty at Shanghai Jiao Tong University, 2023/7-2026/6 (RMB 100,000)
- 2023 Young Investigator Fund of Ruijin Hospital, 2023/11-2024/11 (RMB 30,000)

AWARDS AND HONORS

- 2024 Shanghai Young Investigator Development Program
- 2024 Chinese State Scholarship for post-doctoral fellow
- 2023 Young Investigator Support Program at Shanghai Jiao Tong University School of Medicine
- 2021 Outstanding PhD Graduate of Shanghai
- 2019 National Scholarship for PhD Student
- 2019 Chinese State Scholarship for PhD student
- 2017 National Graduate Mathematical Modeling Competition (Second Prize)
- 2016 Outstanding Undergraduate of Shanghai Jiao Tong University
- 2014 National Biomedical Electronic Innovation Design Competition (Second Prize)

RESEARCH PUBLICATIONS

- **Li Z,** Tu S[*], Matthes M, Li G, Chen Y, Rochitte C, Chen M, Dewey M, Miller J, Scarpa B, Yang W, Qin L, Yan F, Lima J and Arbab-Zadeh A. Prognostic value of coronary CT angiography-derived quantitative flow ratio in suspected coronary artery disease. Radiology, 2024, In Press. (**IF=12.1**)
- **Li Z**[#], Li G[#], Chen L, Ding D, Chen Y, Zhang J, Xu L, Kubo T, Zhang S, Wang Y[*], Zhou X[*] and Tu S[*]. Comparison of coronary CT angiography-based and invasive coronary angiography-based quantitative flow ratio for functional assessment of coronary stenosis: A multicenter retrospective analysis. Journal of Cardiovascular Computed Tomography, 2022;16(6):509-516. (**IF=5.17**)
- **Li Z**, Zhang J[*], Xu L, Yang W, Li G, Ding D, Chang Y, Yu M, Kitslaar P, Zhang S, Reiber J, Arbab-Zadeh A, Yan F and Tu S[*]. Diagnostic Accuracy of a Fast Computational Approach to Derive Fractional Flow Reserve from Coronary CT Angiography. JACC: Cardiovascular Imaging, 2020; 13(1):172-175. (**IF=12.74**)
- **Li Z**, Xu L, Yang W, Li G, Ding D, Chang Y, Yu M, Kitslaar P, Zhang S, Reiber J, Arbab-Zadeh A, Yan F and Tu S. Diagnostic Accuracy of a Fast Computational Approach to Derive FFR from Coronary CT Angiography. Congress of the European Association of Percutaneous Cardiovascular Interventions (EuroPCR) 2019, Paris, France.
- Yang J[#], **Li Z**[#], Tu S[*]. Rationale, validation and development of fractional flow reserve. Chinese Journal of Interventional Cardiology, 2017(8).
- He N[#], Zhang Y[#], **Li Z**, Xu Z, Lyu H, Li J, Dong H, Zhu C, Haacke EM, Mossa-Basha M, Schmidt B and Fuhua Yan[*]. Ultra-High-Resolution Photon-Counting Detector CTA of the Head and Neck: Image Quality Assessment and Vascular Kernel Optimization. American Journal of Roentgenology. 2024. (**IF=4.7**)
- Li G, Weng T, Sun P, **Li Z**, Ding D, Guan S, Han W, Gan Q, Li M. Qi L, Li C, Chen Y, Zhang L, Li T, Chang X, Daemen J, Qu Z and Tu S[*]. Diagnostic performance of fully automatic coronary CT angiography-based quantitative flow ratio, Journal of Cardiovascular Computed Tomography, 2024. (**IF=5.5**)
- Li G, **Li Z**, Chen Y, Chu M, Li C, Wang Z, Yan Y, Luo Y, Cai W, Chen L, Tu S[*]. Atherosclerotic Plaque Characterization by Coronary Computed Tomography Angiography: Comparison With Optical Coherence Tomography. Transcatheter Cardiovascular Therapeutics (TCT) Conference, Journal of the American College of Cardiology, 2023, 82(17):B185.
- Weng T, Gan Q, **Li Z**, et al. Diagnostic accuracy of CCTA-derived versus angiography-derived quantitative flow ratio (CAREER) study: a prospective study protocol. BMJ open, 2022. (**IF=3.007**)

- Guan S, Gan Q, Han W, Zhai X, Wang M, Chen Y, Zhang L, Li T, Chang X, Liu H, Hong W, **Li Z**, Tu S*, Qu X*. Feasibility of quantitative flow ratio virtual stenting for guidance of serial coronary lesions intervention. Journal of the American Heart Association, 2022;11(19):e025663. (**IF=6.107**)
- Westra J, **Li Z**, Rasmussen L, Rasmussen L, Winther S, Li G, Nissen L, Petersen S, Ejlersen J, Isaksen C, Gormsen L, Urbonaviciene G, Eftekhari A, Weng T, Qu X, Bøtker H, Christiansen E, Holm N, Bøttcher M and Tu S*. One-step anatomic and function testing by cardiac CT versus second-line functional testing in symptomatic patients with coronary artery stenosis: head-to-head comparison of CT-derived fractional flow reserve and myocardial perfusion imaging. EuroIntervention, 2021;17(7):576. (**IF=7.728**)
- Qi Y, Xu H, He Y, Li G, **Li Z**, Kong Y, Coatrieux J, Shu H, Yang G* and Tu S*. Examinee-Examiner Network: weakly supervised accurate coronary lumen segmentation using centerline constraint. IEEE Transactions on Image Processing, 2021;30:9429-9241. (**IF=11.041**)
- Zhu L, Pan Z, **Li Z**, Chang Y, Zhu Y, Yan F, Tu S and Yang W*. Can the Wall Shear Stress Values of Left Internal Mammary Artery Grafts during the Perioperative Period Reflect the One-Year Patency? The Journal of Thoracic and Cardiovascular Surgery, 2020;68(08):723-729. (**IF=4.451**)
- Chu M, Gutiérrez-Chico J L, Li Y, Holck E, Zhang S, Huang J, **Li Z**, Chen L, Christiansen E, Dijkstra J, Holm N and Tu S*. Effects of local hemodynamics and plaque characteristics on neointimal response following bioresorbable scaffolds implantation in coronary bifurcations. The International Journal of Cardiovascular Imaging, 2020;36:241-249. (**IF=2.316**)
- Li Y, **Li Z**, Holck E, Xu B, Karanasos A, Fei Z, Chang Y, Chu M, Dijkstra J, Christiansen E, Reiber J, Holm N and Tu S*. Local Flow Patterns After Implantation of Bioresorbable Vascular Scaffold in Coronary Bifurcations - Novel Findings by Computational Fluid Dynamics. Circulation Journal, 2018;82(6):1575-1583. (**IF=2.540**)
- Wu X, Birgelen C, **Li Z**, Zhang S, Huang J, Liang F, Li Y, Wijns W and Tu S*. Assessment of superficial coronary vessel wall deformation and stress: validation of in silico models and human coronary arteries in vivo. The International Journal of Cardiovascular Imaging, 2018;34:849-861. (**IF=1.969**)
- Xu B, Tu S*, Qiao S, Qu X, Chen Y, Yang J, Guo L, Sun Z, **Li Z**, Tian F, Fang W, Chen J, Li W, Guan C, Holm N, Wijns W and Hu S*. Diagnostic Accuracy of Angiography-Based Quantitative Flow Ratio Measurements for Online Assessment of Coronary Stenosis. Journal of the American College of Cardiology, 2017;70(25):3077-3087. (**IF=20.589**)